Day 18 – Test Data & Test Environment

1. What is Test Data?

Test Data is the input data required to test an application or a particular functionality.

Example – Login Page: The username and password entered by a tester are test data. Different inputs help us check whether the login functionality works correctly.

2. Types of Test Data

Type	Meaning	Example
Valid Test Data	Correct input data that should be accepted by the application.	Correct username and password.
Invalid Test Data	Incorrect input data used to check whether the application rejects it or displays a proper error.	Wrong username or password.
Boundary Test Data	Data at or near the minimum and maximum limits of a field.	For an 8–16 character password, test 7, 8, 9, 15, 16 and 17 characters.
Reusable Test Data	Data that can be used again in multiple test cases.	Reuse the same credentials for login testing, retesting and regression testing, when appropriate.

3. Boundary Test Data – Example

Requirement: Password must contain a minimum of 8 and a maximum of 16 characters.

Input length	Expected result	Boundary condition
7 characters	Reject	Just below minimum
8 characters	Accept	Minimum limit
9 characters	Accept	Just above minimum
15 characters	Accept	Just below maximum
16 characters	Accept	Maximum limit
17 characters	Reject	Just above maximum

Note: Boundary testing checks values at the limits and just below or above them. Values 7 and 17 are invalid for this requirement and are also useful boundary test values.

4. What is a Test Environment?

A Test Environment is the setup in which testing is performed. It includes the hardware, software, configuration and tools required to test an application.

Common Components of a Test Environment

• **Application or Website:** The application being tested.

• **Operating System:** Windows, macOS or Android.

• **Browser:** Chrome, Firefox or Edge.

• **Database:** Stores application data.

• **Network:** Internet or required network connection.

• **Testing Tools:** Tools such as Postman or Jira, depending on the testing activity.

Example

To test a website login page, a tester may use a Windows laptop, Chrome browser, internet connection, the test website and valid or invalid login credentials. Together, these form the test setup.

5. Quick Revision

• **Test Data:** Input data required for testing.

• **Valid Test Data:** Correct data expected to be accepted.

• **Invalid Test Data:** Incorrect data used to check validation and error handling.

• **Boundary Test Data:** Data at or near minimum and maximum limits.

• **Reusable Test Data:** Data that can be reused in multiple test cases.

• **Test Environment:** The setup required to perform testing.